

A* Search :-greedy $\rightarrow$ only relying on $h(n)$.

$\hookrightarrow$ The key idea is that, we need to enhance the estimated cost, so that the expensive paths can be avoided or delayed.

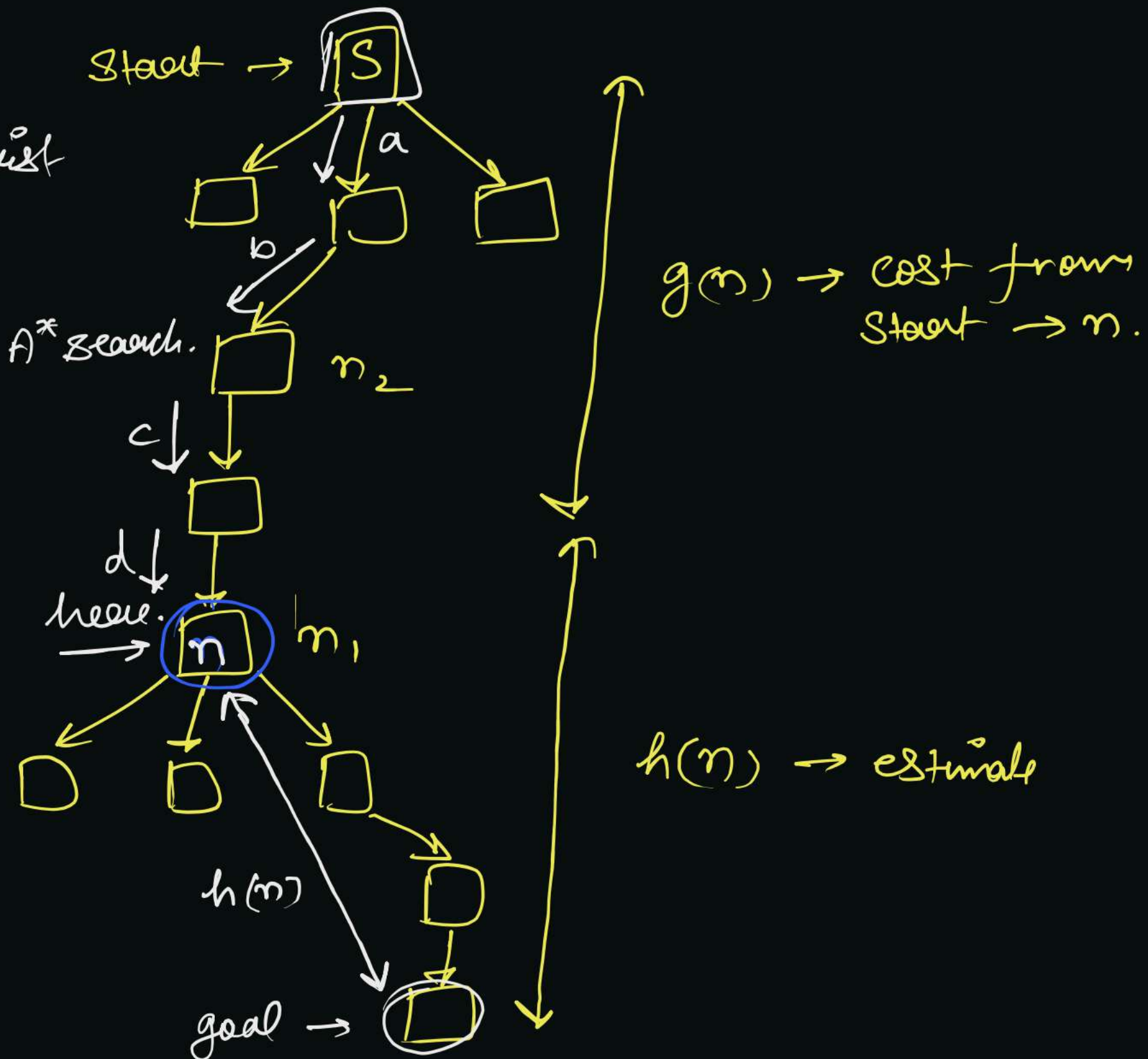
cost $f^n = \boxed{h(n)} \rightarrow$ greedy best-first search.

cost $f^n \Rightarrow \boxed{f(n) = g(n) + h(n)} \rightarrow A^* \text{ search.}$

$\rightarrow$ cost of the path
Start $\rightarrow n$.
 $= a + b + c + d$.

$$g(n_1) > g(n_2)$$

$$\boxed{f(n_1) > f(n_2)}$$



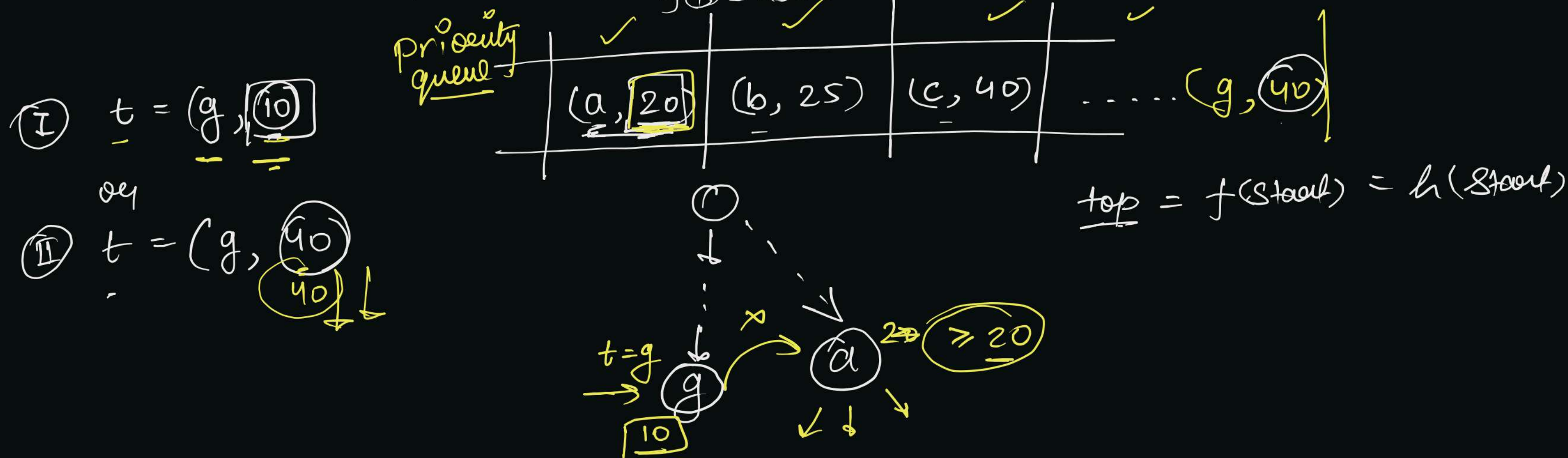
Goal-test :-

↳ In Best-first search,

goal test $\rightarrow$ if $(t \equiv g) \rightarrow$ return t .

↳ In A^* , the goal test : this is to ensure that we reached the goal state : its the optimal one.

if $((t \equiv g) \ \&\& \ (f(t) \leq f(q.top()))) \rightarrow$ return t .



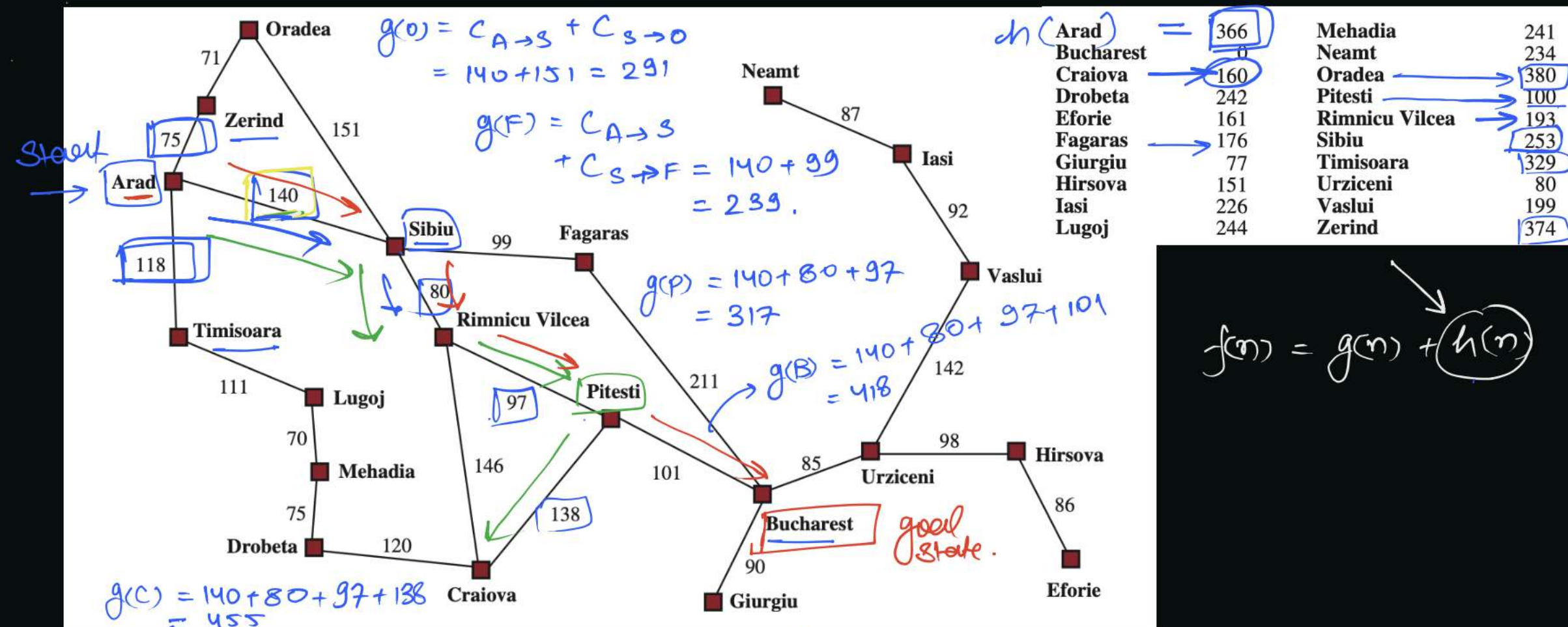
↳ the priority queue will be storing the f -values;

$$f(n) = \underline{g(n)} + \underline{h(n)}.$$

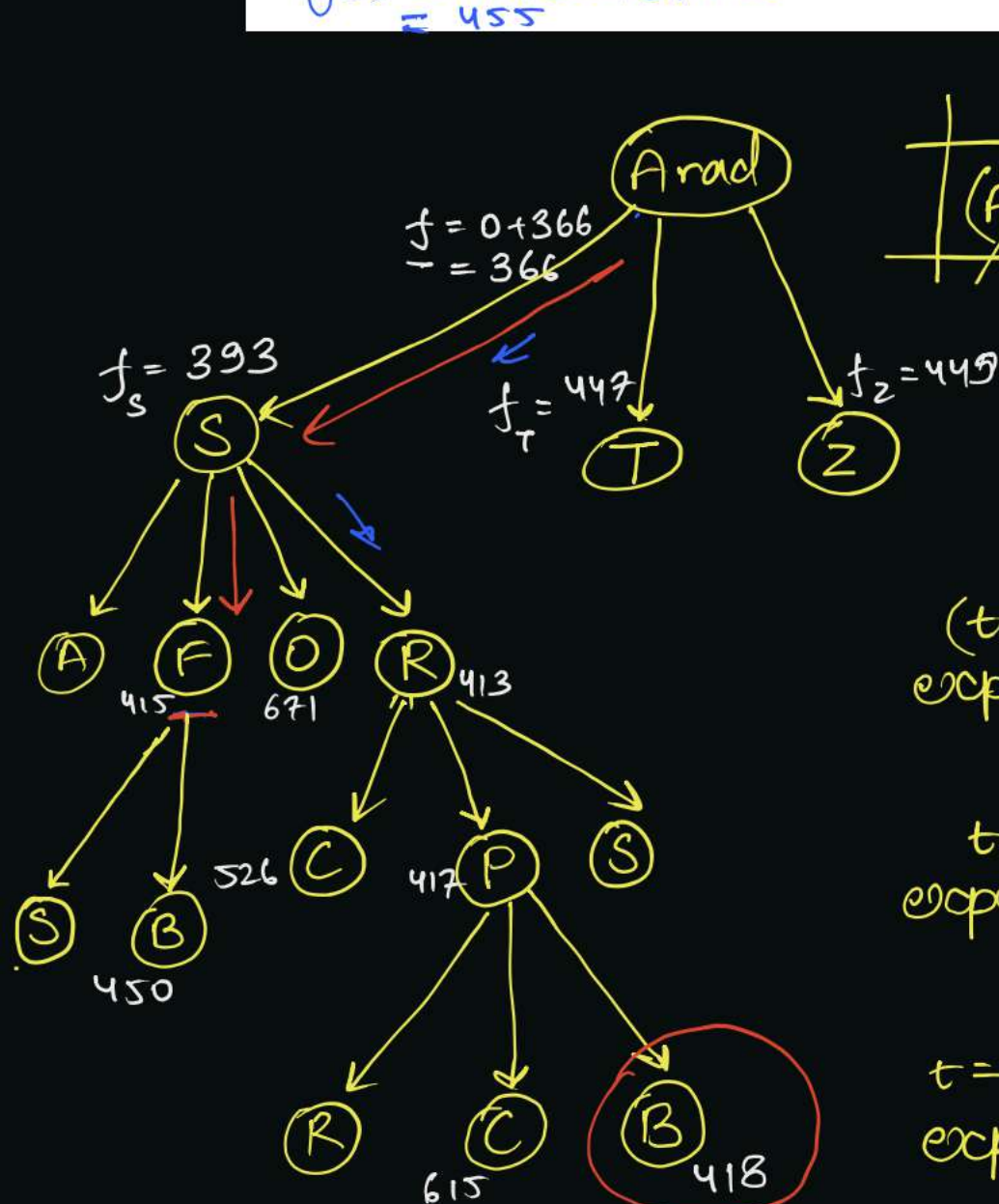
↳ Refining the best looking partial solution until the best solution is refined.

$$f(n) = \underbrace{g(n)}_{\text{uniform cost}} + \underbrace{h(n)}_{\text{heuristic}};$$

A^* algorithm is Dijkstra's algorithm with a sense of direction.



$$f(n) = g(n) + h(n)$$



(A, 366)	(S, 393)	(T, 447)	(Z, 449)	(F, 415)	(O, 671)	(R, 413)	(C, 526)	(P, 417)	(B, 418)	(C, 615)	(B, 418)
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$$f(n) = g(n) + h(n)$$

$$f(A) = g(A) + h(A) = 0 + 366 = 366$$

$$f(t) = f(B) = 418 \downarrow \downarrow$$

$$t = (B, 418)$$

(t = A) expanding (A)

$$\begin{cases} f(S) = g(S) + h(S) = 140 + 253 = 393 \\ f(T) = g(T) + h(T) = 118 + 329 = 447 \\ f(Z) = g(Z) + h(Z) = 75 + 374 = 449 \end{cases}$$

t = S expanding (S)

$$\begin{cases} f(F) = g(F) + h(F) = 239 + 176 = 415 \\ f(O) = g(O) + h(O) = 291 + 380 = 671 \\ f(R) = g(R) + h(R) = 220 + 193 = 413 \end{cases}$$

t = F expanding (F)

$$\begin{cases} f(C) = g(C) + h(C) = 366 + 160 = 526 \\ f(P) = g(P) + h(P) = 317 + 100 = 417 \end{cases}$$

t = B

$$f(B) = g(B) + h(B) = 450 + 0 = 450$$

t = P

$$\begin{cases} f(C) = g(C) + h(C) = 455 + 160 = 615 \\ f(B) = g(B) + h(B) = 418 + 0 = 418 \end{cases}$$

A → S → R → P → B
→ optimal path.

Search terminates because no unexpanded node is having cost less than 418.